RAG Assignment: Baseline Retrieval vs. Cross-Encoder Re-Ranking

1 Setup and architecture

Two retrieval-augmented generation pipelines were compared on the official *CATAN: Cities & Knights* Game Rules & Almanac. A rulebook like this is a demanding test for retrieval. Most of its content is a general rule followed by exceptions, restrictions, and optional variants, so a passage that looks relevant is often not the one that answers the question. The PDF was cut into 118 chunks of up to 160 words with a 40-word overlap, and each chunk kept an ID and its PDF page number.

The models were:

- Embeddings: `sentence-transformers/all-MiniLM-L6-v2`
- Re-ranker: `cross-encoder/ms-marco-MiniLM-L-6-v2`
- Generator: `google/flan-t5-base`

A bi-encoder turns queries and passages into vectors separately, so the passage vectors can be built once and reused for every question. Here they were normalized and kept in a NumPy matrix, which means a simple dot product gives cosine similarity. Exact vector search returned ten candidates, and the baseline handed the top three to the generator.

A cross-encoder takes a different route. The query and a passage go through the model together, and self-attention lets their tokens influence each other before a single relevance score comes out. The advanced pipeline scored the same ten candidates, sorted them by that score, and gave the best three to the generator. Since every score depends on the question, none of it can be precomputed.

Everything else matched across the two pipelines: corpus, initial candidates, prompt structure, generator, deterministic decoding, and a limit of 96 new tokens. The only thing that varied was which passages formed the context and in what order.

2 Qualitative analysis and rank shifts

Seventeen queries were run through retrieval. Three of them also had answers generated by both pipelines: two cases where re-ranking brought in the missing evidence, and one where it did not help. The other fourteen were retrieval-only. The two successful cases were picked after looking through the results and confirming the source pages, which makes them illustrations and rules out reading them as an overall accuracy figure.

2.1 Case 1: Missing discard destination

Question: If I have four progress cards and draw a fifth outside my turn, where must the discarded card go?

Correct rule: Page 9 says a card of the player's choice is discarded to the bottom of the appropriate stack.

Rank shift: `p09_0360` moved from rank 10 to rank 1.

The cause is a chunk boundary. The baseline's top passage, `p09_0240`, states that a card must be discarded and then stops just before saying where it goes. The two other selected passages deal with turn order and with how progress cards are obtained. Re-ranking pulled up the chunk that continues the sentence.

Baseline answer (actual):

```
[p05_0000; page 5] 55 turn o vervie W The player who placed the last city
takes the first turn; afterwards, in clockwise direction, one by one the
other players take their turns.  On your turn, the following actions are
performed, in the order listed:  • Y ou may never hold more than four
progress cards in your hand.  If you draw a fifth progress card, and you
cannot play it because it is not your turn
```

Re-ranked answer (actual):

```
To the bottom of the appropriate stack.
```

The baseline answer copies a stretch of unrelated text and breaks off before reaching anything useful. The re-ranked answer is one short line and correct. Nothing was invented in the baseline output, so the issue is a missing detail that better evidence selection fixed. The cut-off ending also shows how the 96-token limit and the weak extractive behavior of the generator affect results.

2.2 Case 2: Buying improvements without a city

Question: Can I buy city improvements if I no longer have any cities?

Correct rule: Under "Losing Your Last City" on page 12, existing improvements stay in place, but no further improvements can be bought until the player owns a city again.

Rank shift: `p12_0000` moved from rank 10 to rank 1.

Here the baseline fell for shared vocabulary. It retrieved the ordinary improvement rules and a restriction tied to metropolises. Both mention improvements and cities, yet neither says what happens with zero cities. The re-ranked context contained the passage written for exactly that situation.

Baseline answer (actual):

```
You may not purchase any improvements beyond the third level of a given
color unless you have a city where you could build a metropolis.  If you do
not have such a city on the board, you must wait until you have built at
least one more city.
```

Re-ranked answer (actual):

```
Once you own a city again, you may continue to purchase city improvements
normally.
```

The baseline answer describes a limit that applies above level three, so purchases at lower levels with no cities stay unanswered. The re-ranked answer speaks to when buying resumes. It is still short on clarity. A direct statement that purchases are blocked until a city is owned again would read better, and no passage is cited.

2.3 Regression diagnostic

For the question about activating a knight and using it against the robber in the same turn, re-ranking moved `p13_0480` from rank 8 to rank 1. That passage covers an optional pre-roll variant, a different timing issue from the normal activation restriction on page 6. Both generated answers talk about chasing the robber away and neither settles whether the knight may act right after activation. The result is a retrieval regression. It shows that a cross-encoder can favor a closely related optional rule and still miss the condition the question depends on.

3 Latency profiling

The run used Windows 11 and Python 3.12.5. Generation ran on the CPU with four PyTorch threads configured, and the exact CPU model was not recorded. Models were warmed up before timing, and each query was timed five times. The values below are medians from the saved `experiment.json`.

Query	Baseline (ms)	Re-ranking (ms)	Advanced (ms)
Discard destination	37.54	640.39	694.73
No cities remaining	16.41	462.97	483.05
Knight activation	18.65	468.96	487.97

Table 1: Median retrieval latency over five repetitions.

Baseline time covers query embedding and exact vector search. The advanced figure adds cross-encoder scoring and sorting. Model loading, PDF extraction, indexing, and generation are left out. Because medians are taken per component, they do not have to add up exactly to the median total. Embedding all passages and building the index took roughly 6.27 seconds according to the saved metadata.

Using unrounded values, advanced retrieval was about 18.5 times slower for the discard question and 29.4 times slower for the no-cities question. In both examples the extra time bought the missing evidence, though that says little about how often it would do so on other queries.

Query	Baseline generation (ms)	Re-ranked generation (ms)
Discard destination	17,925.04	3,096.63
No cities remaining	16,142.32	4,717.03

Table 2: Answer generation time (single observations).

Generation times come from single runs and are not five-run medians. The re-ranked answers were much shorter, and fewer output tokens almost certainly account for a good part of the

lower times. A general claim that re-ranking speeds up the full system cannot be drawn from two measurements like these.

4 Why not cross-encode the entire collection?

A cross-encoder has to read each query together with each candidate passage. Scoring a whole collection of N passages therefore means N query–passage pairs for every question, and batching only makes that work run faster without reducing it. Dense passage embeddings avoid the problem because they are computed once and shared across all queries.

Two-stage retrieval keeps the expensive scoring to ten pairs. For this corpus that is 10 pairs instead of 118, or $118/10 = 11.8$ times fewer. That ratio counts pairs only and was not measured as a speedup. For much larger collections the gap grows, and the first stage can also use an approximate nearest-neighbor index to stay fast.

The price is candidate recall. A passage that never makes the initial top ten cannot be rescued by re-ranking, so the first-stage retriever has to deliver a candidate set that actually contains the answer.

Source: Official *CATAN: Cities & Knights* Game Rules & Almanac, especially pages 9 and 12.
Repository: `https://github.com/Tejaswini-123-mannem/rag-baseline-vs-reranking`